

LANDON BAKKEN

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Education

University of Wisconsin Madison

Bachelor in Computer Engineering, expected July 2028

Sept 2024 – Present

Madison, Wisconsin

Relevant Coursework: ECE 252, 352, 210, 230, 354 — CS 240, 300, 400

Experience

Wisconsin Racing Formula SAE

Sept 2024 – Present

Controls Lead

Madison, Wisconsin

2026 Electric Car

- Created software, driver, and physics in loop simulation that contributed to **1st place in Overall Design**, reducing on-track tuning/development time by **64.4%**, and allowed for **80%** more control systems developed
- Developed unconventional transient traction control system, reduced slip RMSE by **29.2%**, resulting in **3rd place Autocross**. Wasted wheel slip energy was reduced by **71.03%** compared to no traction control, contributing to **5th place Efficiency**
- Developed launch control, improved average acceleration by **8.5%**, contributing to **1st place Acceleration**
- Made four wheel drive speed estimation system that has a normalized RMSE of **3.43%** during autocross

2025 Electric Car

- Developed and tuned traction control with load transfer feedforward and slip error feedback controller - reduced lap times resulted in **2nd place Autocross**
- Developed and tuned torque vectoring based on speed and steering angle. Recognized by the **Multimatic Vehicle Dynamics Award**
- Developed driver interface relating charge and distance left in endurance that was specifically recognized for **1st place in Overall Design**
- Managed a multi-bus CAN communication system across 20+ devices, including inverters, ECU, BMS, datalogger, and sensors

Combustion Car

- Developed systems to tune engine cylinder air to fuel ratio, resulting in a **13.2%** power increase, and **3rd place in Efficiency**
- Made torque model that allows for precise and immediate torque cuts for traction control, faster pneumatic shifting, and faster throttle response, contributing to **3rd place in Overall Design**

Mercury Marine

May 2026 – Present

Embedded Controls Engineer

Oshkosh, Wisconsin

- Removed fuel flow sensor noise due to booster pump overshoot, reduced peak-to-peak noise by **45%**
- Developed an autonomous navigation augmentation enabling pre-set pattern execution (figure-8s, sweeps), reducing design cycle time by **1 year**
- Built a hull simulator visualizer allowing developers to visually verify motor collisions in simulation, accelerating development

Personal Projects

Machine Learning | *Neural Networks, Gradient Descent, Python*

Dec 2024 – Jan 2025

- Built a machine learning model and training system from scratch in Python, replicating mathematical formulas and learning to play games including Flappy Bird and Connect 4

Multiplayer Networking | *Networking, C#, Unity*

Nov 2023 – Dec 2024

- Developed a low-latency, peer-to-peer multiplayer system for Unity using UDP, TCP, and HTTP protocols

Technical Skills

Computer Languages: Simulink, Python, C#, Java, C++, Javascript, Verilog, CSS/HTML

Manufacturing: SLA and FDM 3D printing, Laser Cutting, Soldering, CNC

Development Tools: Git, VS Code, Kvaser, Unity, SVN

3D Design: Blender, Fusion, Solidworks

Concepts: PID Controllers, CAN Protocol, Traction Systems, Signal Filtering, Networking, Vehicle Dynamics, Machine Learning